Monetary Policy and Central Bank Information in Colombia

Daniel Osorio and Oscar Jaulín-Mendez

Banco de la República de Colombia

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Motivation

- Information Channel has been an important topic in applied maroeconomics during the last decade.
- HFI of monetary policy shocks still contain central bank information.
- **Consequence:** Price Puzzle.
- identification using stock markets and financial markets ([**?**] [**?**])
- Few evidence for emerging economies. No evidence for Colombia.

Information channel in Colombia

Stock market IS NOT informative in Colombia

- Stock market highly respondent to exogenous variables (Oil prices).
- Using it lead to misleading conclusions.

Break Even inflation IS informative in Colombia

- BEI measure is robust to different information from the central bank.
- Shows to be robust in identifying correctly the CBI channel.

Related Literature

- **Information in MPS:** Miranda-Agrippino (2016), Cieslak and Schrimpf (2018), Nakamura and Steisson (2018), Jarocinski and Karadi (2020), Bauer and Swanson (2023).
- **High Frequency in EMEs:** Kohlscheen (2014),Pirozhkova et al. (2024), Checo et al. (2024), Bolhuis et al (2025).
- This paper: Extract central bank information from monetary policy surprises using BEI.

Outline

1. Identification
2. Estimation
3. Results

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1 Identification

2 Estimation

3 Results

Identification

- Let S_t be the M.P interest rate minus what it is expected by the market (OIS, bloomberg).
- Financial instrument:
  1. Break Even Inflation.
- Changes around meetings (one day after vs one day before) have different information:
  1. **Increase:** Higher inflationary pressures (i.e higher expected inflation).
  2. **Decrease:** Smaller inflationary pressures (i.e. smaler expected inflation).

Identification

Model Diagram.PNG

Figure: Scatter plot of S_t and changes in BEI around monetary policy announcements.

Identification

- What part of the unanticipated surprises in interest rates are related to increases in expected inflation?
- Jarocinski and Karadi (2020) use sign restrictions:

Variable	u_t^M	u_t^I
S_t	$+$	$+$
ΔBEI_t	-	$+$

- u_t^M: raw monetary policy shock. The part of the S_t with decrease in BEI.
- u_t^I: CB's information. The part of the S_t with increases in BEI.
- The procedure guarantee that $S_t = u_t^M + u_t^I$.

Estimation

Local Projection

Asume we can identify perfectly the shock. Then the IRF of the CPI can be estimated using:

$$P_{t+h} - P_{t-1} = \alpha_h + \beta_h S_t + \varepsilon_{t,h}$$

Shocks can be correlated with previous macroeconomic data (Bauer and Swanson, 2023):

$$P_{t+h} - P_{t-1} = \alpha_h + \beta_h S_t + X_t \gamma_h + \varepsilon_{t,h} \tag{1}$$

For $\hat{\beta}_h$ to be consistent, we need:

$$E[S_t, \hat{\varepsilon_{t,h}}] = 0$$

Identification

LP

The central bank information implies:

$$E[S_t, \hat{\varepsilon_{t,h}}] \rightarrow E[S_t, I_t + \varepsilon_{t,h}]$$
$$E[S_t, I_t] + E[S_t, \varepsilon_{t,h}] \rightarrow E[S_t, I_t] \neq 0$$

If I_t is inflationary $\rightarrow$ there exists price puzzle.
Identification assumption:

$$E[u_t^I, I_t] \neq 0$$
$$E[u_t^M, I_t] = 0$$

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Empirical results

LP

We estimate IRFs with LP in equation (1) using as shocks $Z_t \in \{S_t, u_t^M, u_t^r\}$. X_t contains CPI, GDP an interest rate.

- S_t: Price puzzle.
- u_t^r: Price puzzle driven by CBI.
- u_t^M: No price puzzle.

IRFs

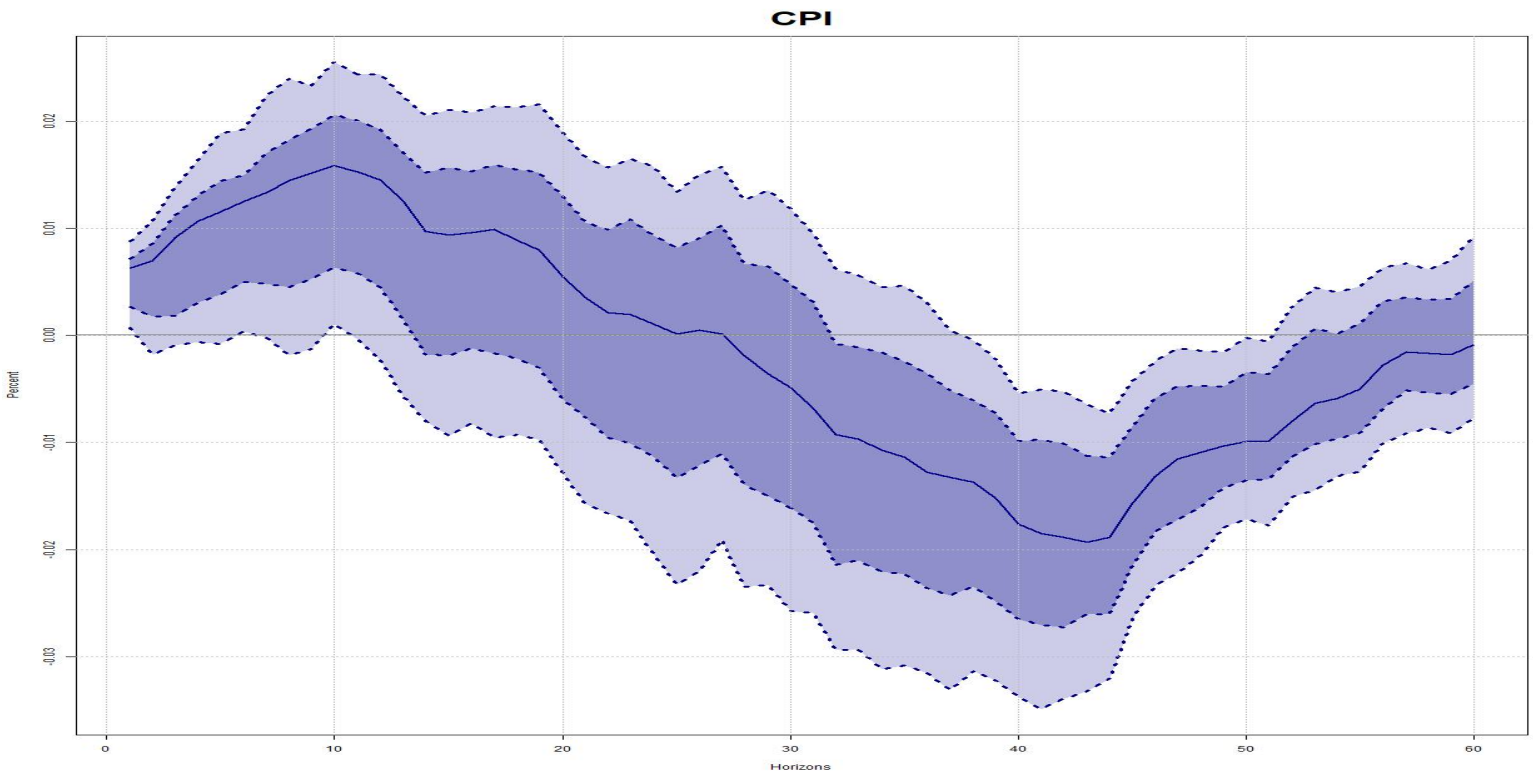


Figure: IRF to a shock S_t. Confidence intervals at 68.0% and 90.0%.

IRFs

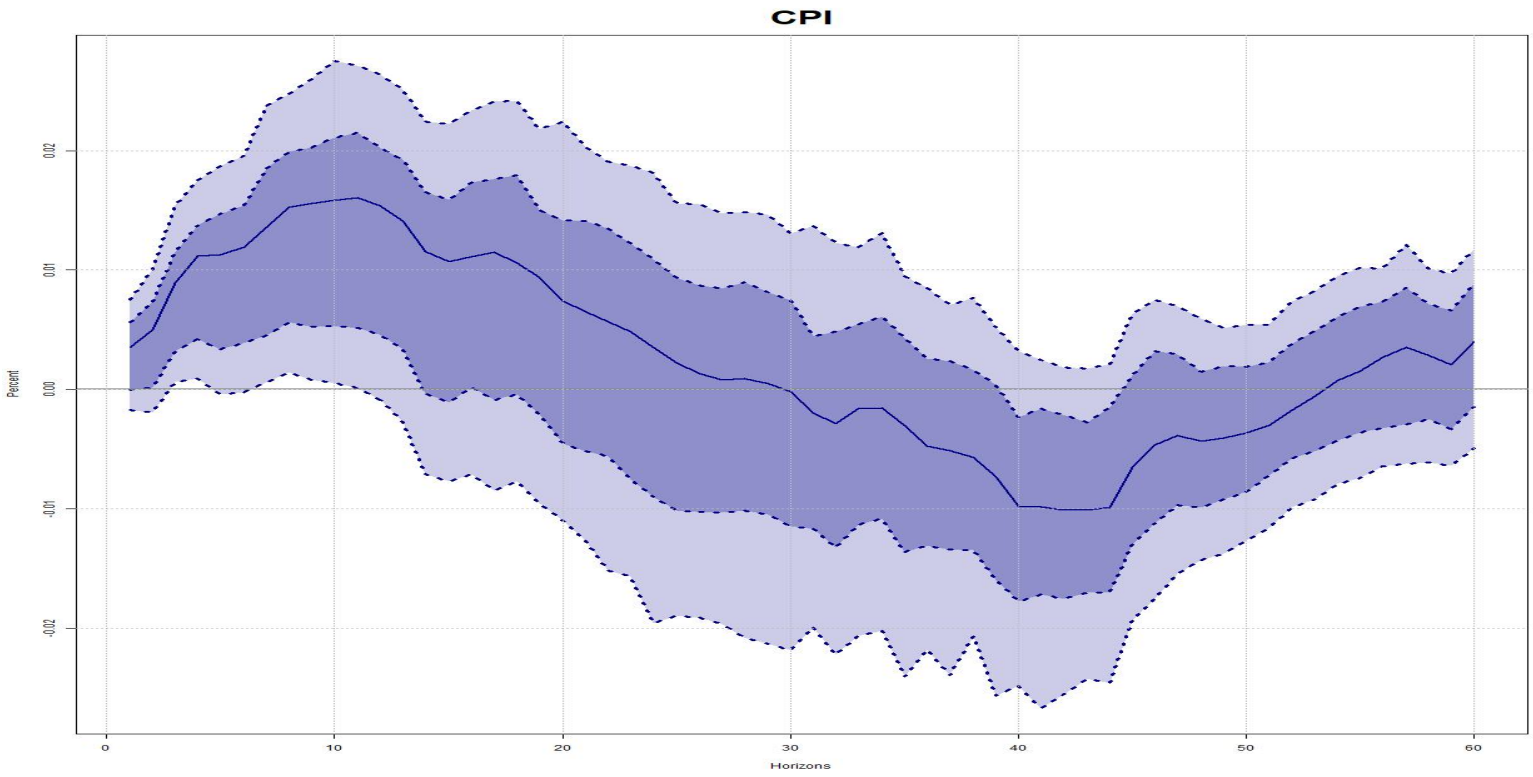


Figure: IRF to a shock u_t^r. Confidence intervals at 68.0% and 90.0%.

IRFs

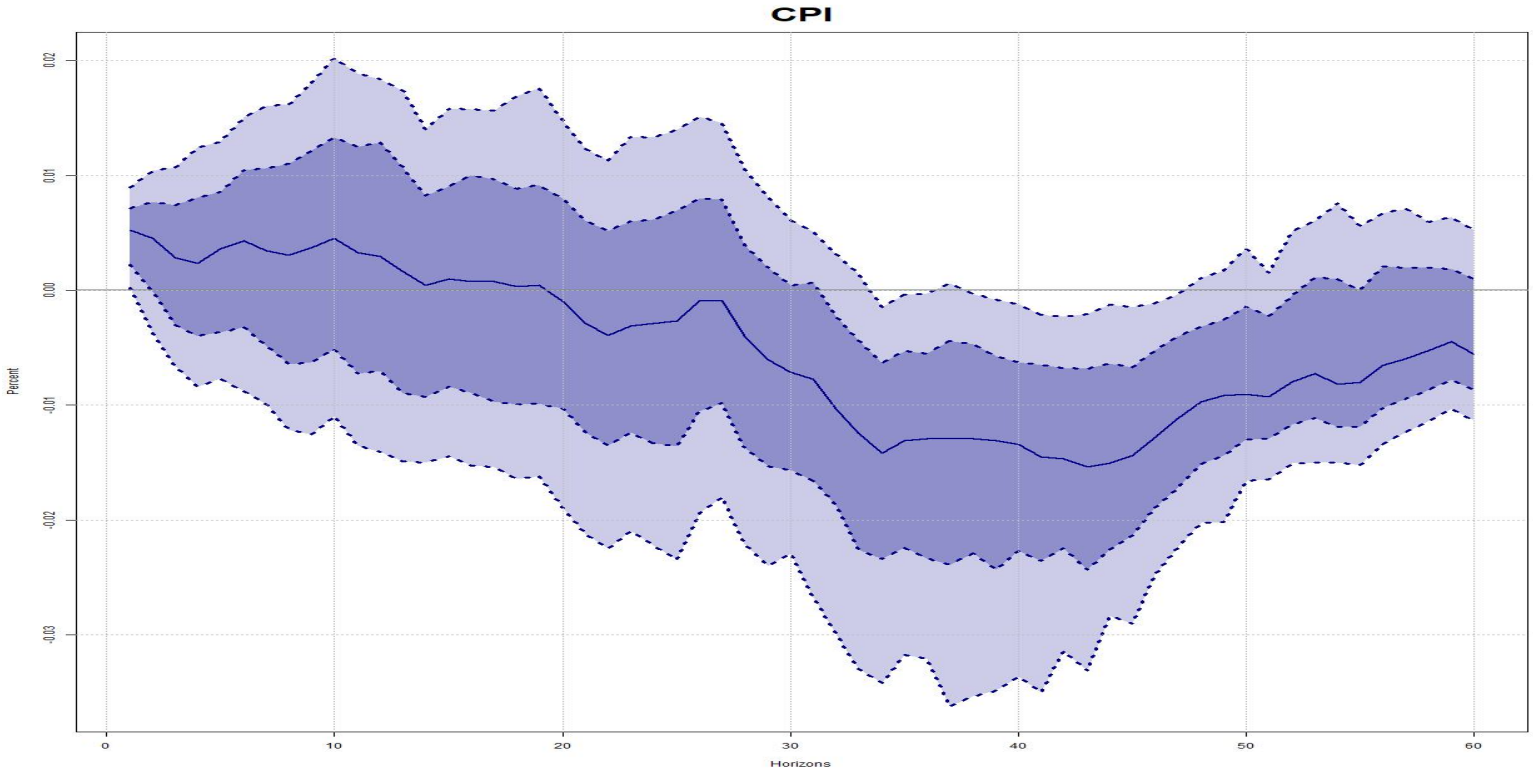


Figure: IRF to a shock u_t^M. Confidence intervals at 68.0% and 90.0%.

Bloomberg Expectations

- S_t: Decrease output and price puzzle.
- u_t^r: Price Puzzle driven by CBI.
- u_t^M: Decrease output AND reduction in price level (No price puzzle).

IRFs

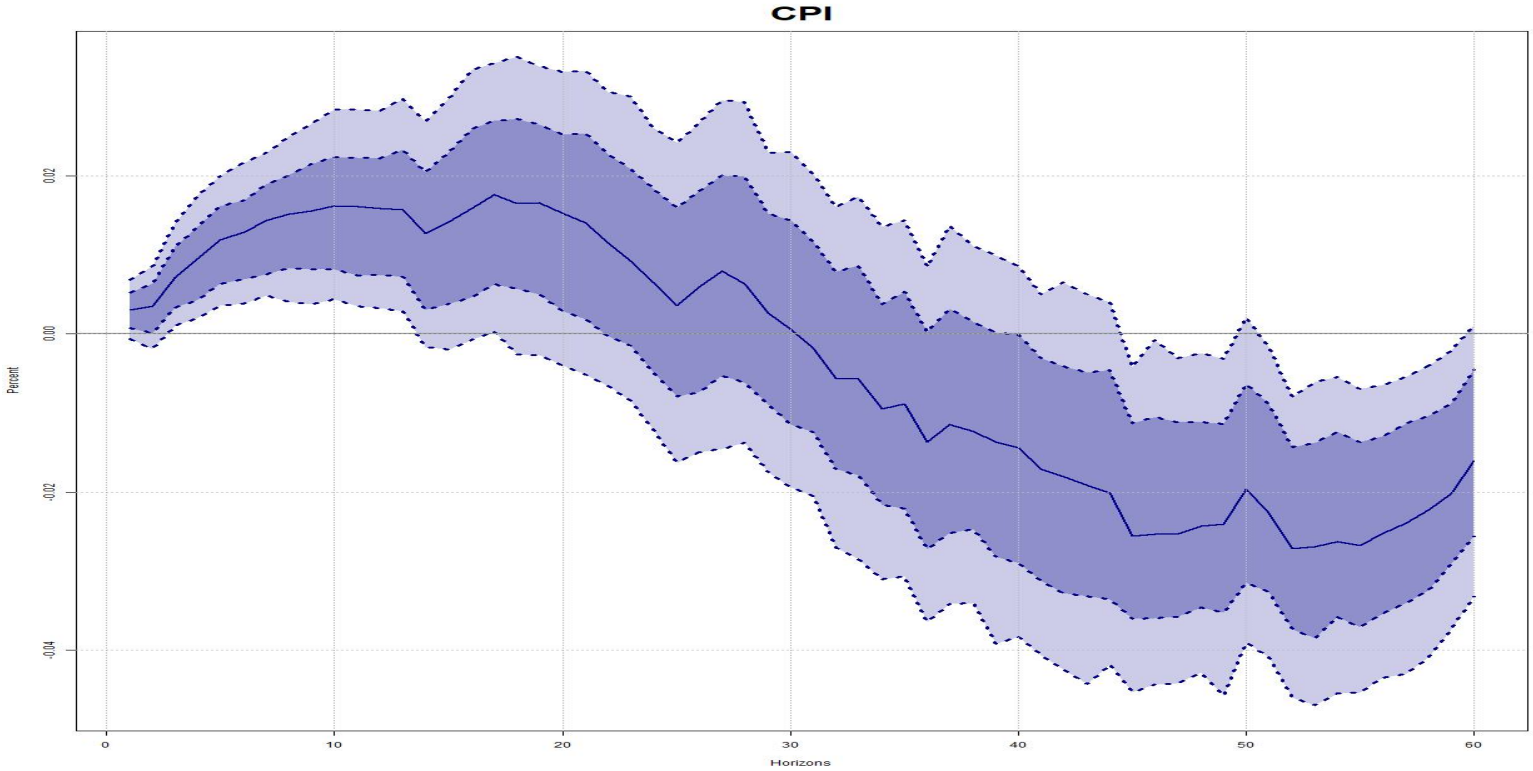
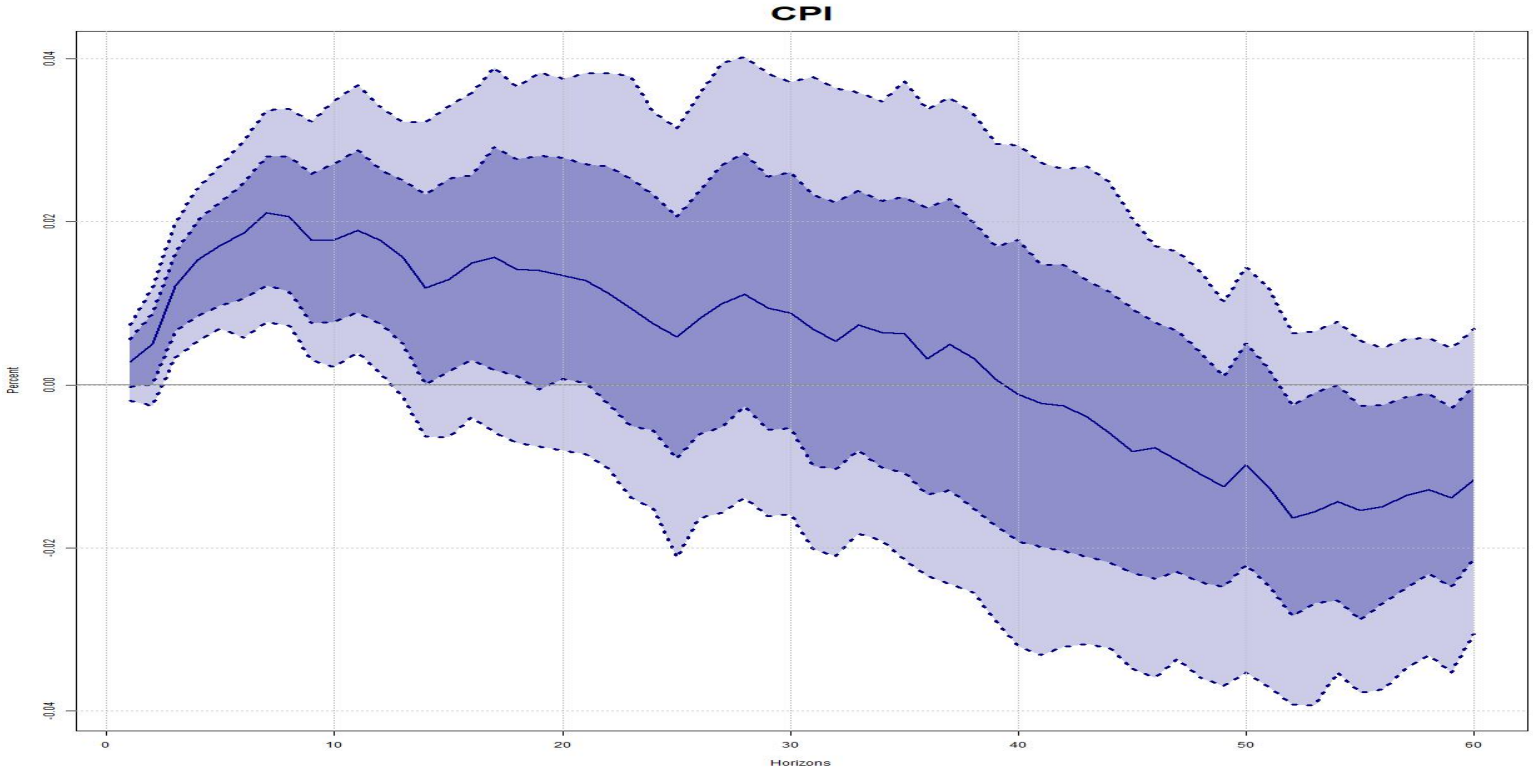


Figure: IRF to a shock S_t using Bloomberg for expectations. Confidence intervals at 68.0% and 90.0%.

IRFs



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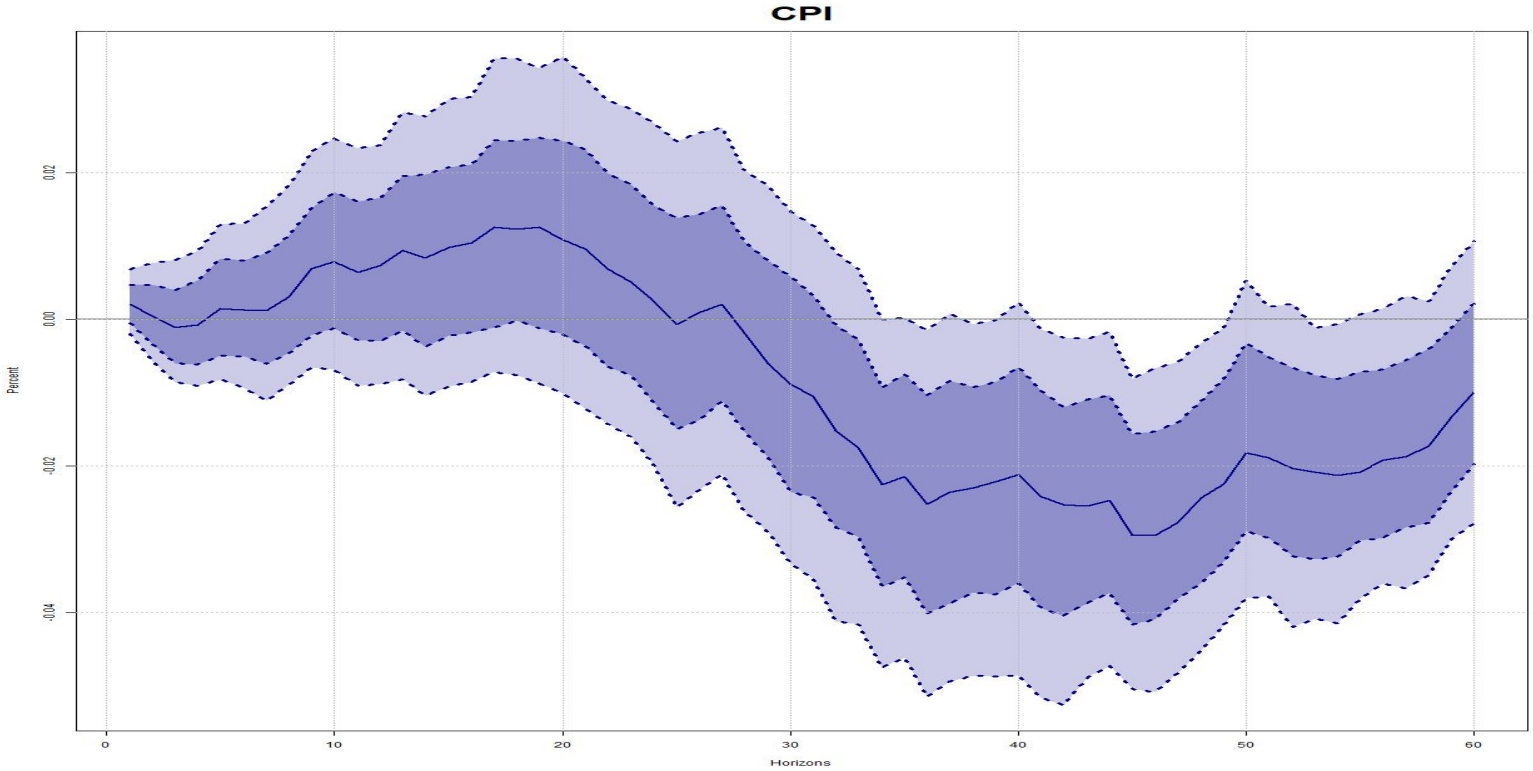

Figure: IRF to a shock u_t^M using Bloomberg for expectations. Confidence intervals at 68.0% and 90.0%.

Stock markets?

The conclusion might be different:

- u_t^r: Decrease in prices.
- u_t^M: Price Puzzle.

IRFs

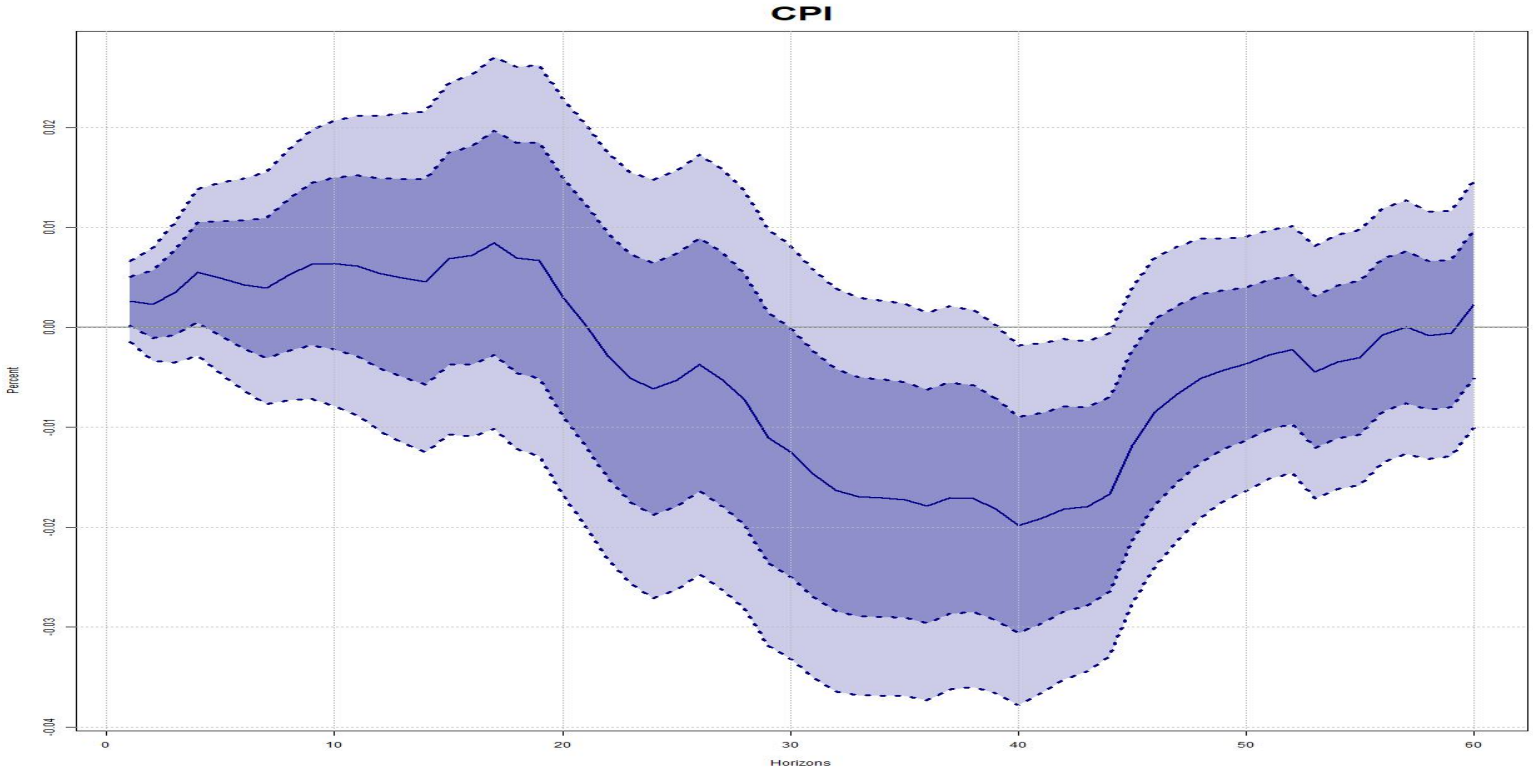

Figure: IRF to a shock u_t^r using COLCAP to orthogonalize shocks. Confidence intervals at 68.0% and 90.0%.

IRFs

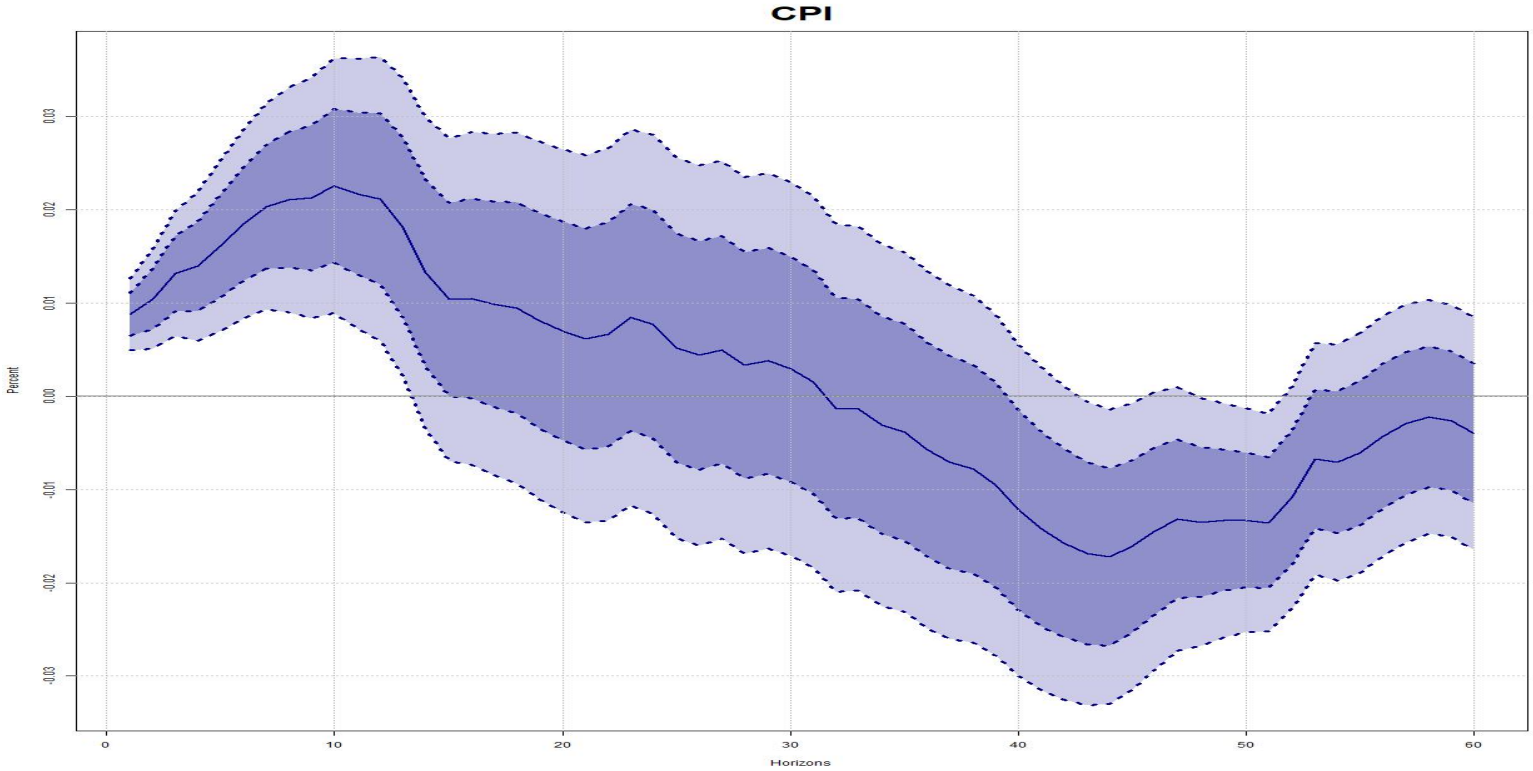

Figure: IRF to a shock u_t^M using COLCAP to orthogonalize shocks. Confidence intervals at 68.0% and 90.0%.

Robustness

1. Results are very similar if we run a BVAR.
2. Orthogonalizing the shocks an the using them in the LP is very similar.